

Indicators of sustainable forestry: The association between wildlife species and forest structure in Finland

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ABSTRACT

There is a long tradition of sustainable forest policy in Finland. Today, forests should be managed on the basis of the principles of sustainable forestry. However, economics have driven such forestry practices, resulting in dramatic changes to the natural structure and dynamics of Finnish forests. Commercial forests should, however, also provide habitats for various organisms, and forestry activities should not reduce forest biodiversity. Indicators can provide useful information on the ecological status and trends in sustainable forestry, such information can then be used by decision-makers. In attempts to develop accurate and reliable indicators, it has been suggested that statistical data, inventory data and data on many taxa may prove useful. The richness of game species and the abundance of the Siberian flying squirrel have been considered as candidate indicators for sustainable forestry. Indicators of sustainable forestry are commonly used separately and the interrelations between the ecological, economic and social dimensions of forestry have rarely been studied. In this paper we examine associations between the abundance of animal species (16 game species and Siberian flying squirrels) and their co-occurrence with forest structural characteristics that are considered to be indicators of sustainable forestry. We were also interested in examining regional patterns of the variables, and determining which socio-economic indicators and land-use factors were associated (positively or negatively) with areas of forest for which there were indicators suggesting a good ecological condition. The study area consisted of 39 observation units. Based on our exploratory and descriptive analyses we suggest that using indices for different wildlife species groups and relative abundances of the Siberian flying squirrel together with forest structural characteristics and adequate socio-cultural variables could provide relevant information about trends in sustainable forestry. The application of indicators always depends on socio-cultural contexts and biogeographical factors, and these must be taken into consideration when making predictions or comparing different regions.

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1. Introduction

Sustainable forest policy has a focal position in Finnish decision-making; the traditional goal has been to safeguard continuous timber supplies (Vierikko et al., 2008). Today forests should be managed according to the principles of sustainable forestry that link economic, ecological, social and cultural aspects on an equal footing (Ministry of Agriculture and Forestry, 2008).

The implementation of economic and ecological sustainability goals is a challenging task, especially in Southern Finland, because of the structure of forest ownership where 73% of forests are owned by non-industrial private forest owners, and the long land-use history compared to that in Northern Finland. In addition, the proportion of strictly protected forest land is much lower in the South than in the North: 2.2% and 20.2%, respectively (Finnish Statistical Yearbook of Forestry, 2005; Vehkamäki et al., 2006).

Indicators provide useful information for decision-makers about the status of and trends in commercial forests, supplying evidence about whether economic, ecological and social targets and goals for sustainable forestry have been achieved (Larsson, 2001; Gough et al., 2008). In particular, administrators seek

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information that will aid them in the implementation of political goals such as sustainable development, i.e., in steering forest-related actors, their actions and practices (Gustavson et al., 1999; Vierikko et al., 2008). For example, criteria and indicators have been developed for assessing and monitoring sustainable forestry in Europe and Finland (Ministry of Agriculture and Forestry, 2007). These indicators are based on available statistical data such as the extent of conservation areas and the number of employees in the forestry sector. Structural characteristics of forests (e.g. tree species composition, tree age structure) that are collected during the national forest inventories are commonly used indicators of sustainable forestry. Land-use history and land-ownership can be valuable indicators when monitoring sustainable forestry. Larsson (2001), for example, suggested that forest ownership could be a key factor in European forest biodiversity at national and regional scales. Several species have been proposed as candidate indicators in boreal forests. Epiphytic lichens, wood-rotting fungi and wood inhabiting invertebrates have been suggested as suitable indicators of forests of high conservation value (e.g. Tibell, 1992; Kuusinen, 1996; Kuusinen and Siitonen, 1998; Martikainen et al., 2000; Ihlen et al., 2001).

Studies of game animal assemblages have provided support for their use as indicators of sustainable forestry and land-use changes to assess, e.g. forest fragmentation at the landscape scale (Pellikka et al., 2005, 2006; Edman, 2008). For example, grouse and particularly the capercaillie are considered good candidate “umbrella” species and indicators of the diversity of many other bird species and wildlife richness in the boreal forest landscape (Pakkala et al., 2003). Abundance data for game animals are valuable, because their habitats are well-known, and they are ecologically, socially, culturally (even spiritually) and economically valuable species. Therefore, information about game animals as indicators may be strongly linked to aspects of sustainable forestry in several contexts.

Another potential indicator of sustainable forestry is the vulnerable, arboreal Siberian flying squirrel (*Pteromys volans*).

The population size and habitat requirements of the species are relatively well-known. Earlier studies have suggested that the flying squirrel could be an indicator of diverse forest structure and, consequently, of the presence of other forest species requiring mature forest landscapes (Hurme et al., 2008). Hurme et al. (2008), however, studied the flying squirrel at its northernmost distributional limits in Finland, and their results cannot necessarily be extrapolated to the main distribution area of the species. Game species and the flying squirrel share many characteristics that make them interesting from a decision-making point of view. They can easily be regarded as charismatic flagship species with high public and political interest (Shrader-Frechette and McCoy, 1993).

Developing accurate and precise indicators of sustainable forestry has turned out to be a challenging task. Indicators should, for example, have high policy relevance, be easy to interpret, enable forecasting and comparison and be available at a reasonable cost (Rosentröm, 2009, p. 17). A problem associated with practical applicability, for instance, is that actions taken by administrators operate at different spatial and temporal scales than ecological processes (Cumming et al., 2006; Tasser et al., 2008). A common problem in developing indicators for setting up goals for sustainable forestry is that economic, social and ecological indicators are traditionally examined separately without analyzing their interrelations. However, there are several studies concerning the interrelationships and co-occurrence of different taxonomic groups and their responses to forestry activities at the landscape or broader regional scale (e.g. Nilsson et al., 1995; Lindenmayer and Cunningham, 1997; Lawton et al., 1998; Martikainen et al., 1998; Pharo et al., 1999; Caro, 2001; Sverdrup-Thygeson, 2001; Bani et al., 2002; Bayne et al., 2004; Roberge and Angelstam, 2006; Similä et al., 2006; Hurme et al., 2008).

The aim of this study was to shed light on the association between potential indicators of sustainable forestry and data sources relating to wildlife species and forest structural characteristics. We wished to examine the value of combining various indicators of sustainable forestry with land-use and socio-

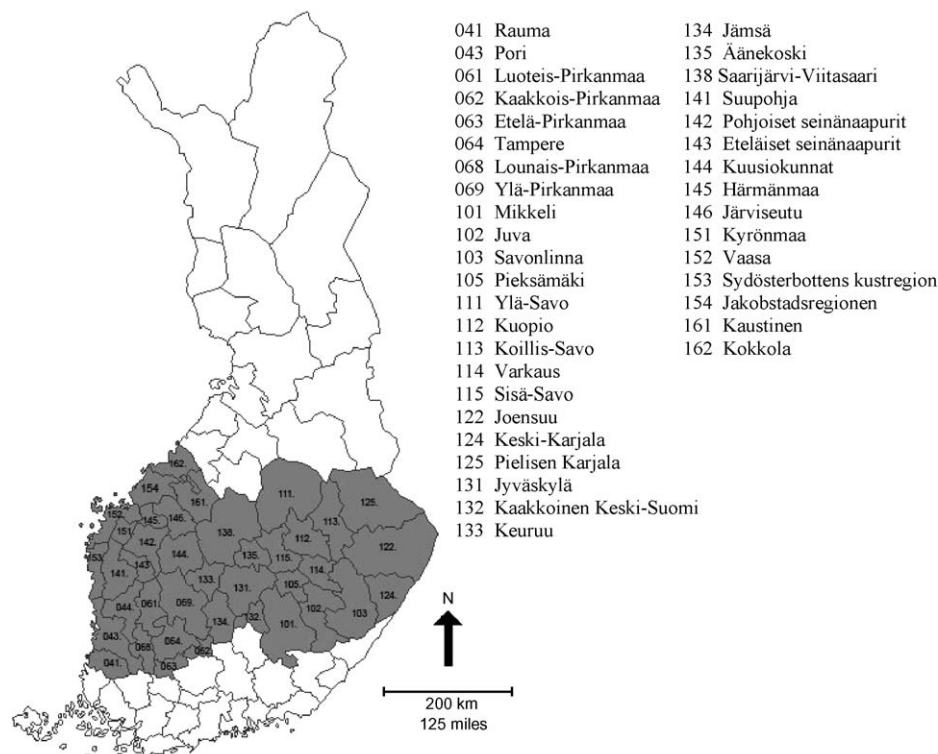


Fig. 1. The codes and names of the 39 regional municipality complexes, i.e. observation units, situated in the study area.

economic variables (land-ownership, economic and forestry activity, number of inhabitants). In our preliminary study we focus on the similarities and differences in the information content of indicators and make use of comparative descriptions. More specifically, in this paper we: (1) examine associations between the abundances of wildlife species (game animals and the Siberian flying squirrel) and forest structural characteristics; (2) describe regional distributions of species and forest structural characteristics; and (3) examine how the indicators of ecological status are associated with socio-economic indicators and land-use factors reflecting other dimensions of sustainable forestry.

2. Materials and methods

2.1. Study area

The study area consisted of 39 observation units, i.e. LAU-1 (Local Regional Units 1), ranging in size from 86,000 to 793,200 ha; these were located in central Finland (Fig. 1) (European Commission, 2008). The practical advantage of using these units is that they are commonly used in recording the economic and social phenomena of regional communities. In addition, this scale is sufficiently detailed to detect relative changes in the richness and abundances of wildlife populations (Vierikko et al., 2008).

Biogeographically, the study area is located in the southern and central boreal vegetation zones (Ahti et al., 1968). Forest land in Finland is classified on the basis of its productivity, i.e., annual increment of the growing stock: productive forest land ($\geq 1 \text{ m}^3/\text{ha/a}$); low productivity forest land ($0.1\text{--}1 \text{ m}^3/\text{ha/a}$) and waste land ($<0.1 \text{ m}^3/\text{ha/a}$). The amount of forest land in the 39 observation units ranged from 61,514 to 698,196 ha and the proportion of productive forest land varied between 79.5% and 99.2%, with the lowest proportion in the western part of the study area. Almost all of the productive forest land in the study area is commercially managed (ca. 99%) (Finnish Statistical Yearbook of Forestry, 2005).

2.2. Species data

The distribution patterns and relative abundances of the Siberian flying squirrel were derived from 6487 plots (study plot

size 9 ha) in the study area during 2003–2005 (Hanski, 2006). The presence of fecal pellets was used to indicate the presence of flying squirrels. The Siberian flying squirrel population size was calculated on the basis of data from the occupied study plots and the portion of the total area examined (Hanski, 2006). In this study, we used the proportion of occupied study plots per regional municipal complex as an observation unit for the analysis (Table 1).

The wildlife richness index (WRI) was developed by the Finnish Game and Fisheries Research Institute; it uses wildlife triangle counts (Lindén et al., 1996). Data were collected by volunteer hunters, who recorded information on the abundances of 30 species. The data used consist of abundance estimates (animal tracks in the snow per 10 km of census transect per 24 h, or game birds observed within the census transect belt) for the 16 wildlife species recorded within the observation units during 1995–2004, compared to average abundances in the whole study area during 1989–1994. The indices for each observation unit (g) were calculated using the following formula:

$$\text{WRI}_g = \sum_{i=1}^S \log \left(\frac{a_{it}}{R_{iT}} + 1 \right)$$

where WRI_g is the wildlife richness index in observation unit g , S is the number of species considered, a_{it} is the abundance of species i in the time period t in unit g and R_{iT} is the abundance of the same species during time period T in the reference area (Pellikka et al., 2005). The WRI indices were calculated separately for five wild animal groups: large carnivores, small carnivores, ungulates, grouse and other species (Table 1).

2.3. Forest structural data

Tree and forest land data were collected from permanent systematic cluster samples during the period 1996–2003 (9th National Forest Inventory of Finland). There were about 70,000 systematic cluster samples in Finland, covering more than 200,000 ha of land (sampling error for the total volume of the growing stock is less than 1.0%). Data were derived from the operative multiple-source inventory system, which utilized field

Table 1
Summary statistics relating to the Siberian flying squirrel and wildlife richness indices for various species groups.

Variable	Species list	Name	Min.	Max.	Mean	SD
Relative abundance of Siberian flying squirrel		Ptevol	3.73	47.5	14.05	10.45
Wildlife richness index (16 wildlife species)		WRI_total	7.37	17.03	12.29	1.94
Wildlife richness index of four game animal groups	WRI-grouse (4)	WRI_grouse	2.53	4.72	3.57	0.53
	Black grouse (<i>Tetrao tetrix</i>)					
	Capercaillie (<i>Tetrao urogallus</i>)					
	Hazel grouse (<i>Bonasa bonasia</i>)					
	Willow grouse (<i>Lagopus lagopus</i>)					
	Large carnivores (3)	WRI_large	0	6.00	1.61	1.41
	Lynx (<i>Lynx lynx</i>)					
	Wolf (<i>Canis lupus</i>)					
	Wolverine (<i>Gulo gulo</i>)					
	Small carnivores (3)	WRI_small	1.83	4.08	2.80	0.54
	Pine marten (<i>Martes martes</i>)					
	Red fox (<i>Vulpes vulpes</i>)					
	Stoat (<i>Mustela erminea</i>)					
	Ungulates (4)	WRI_ungu	0.61	5.32	2.18	1.37
	Moose (<i>Alces alces</i>)					
	Roe deer (<i>Capreolus capreolus</i>)					
	White-tailed deer (<i>Odocoileus virginianus</i>)					
	Wild forest reindeer (<i>Rangifer tarandus fennicus</i>)					
	Other forest species (2)	WRI_other	1.59	4.00	2.13	0.49
	Mountain hare (<i>Lepus timidus</i>)					
	Red squirrel (<i>Sciurus vulgaris</i>)					

Table 2

Summary statistics for the forest structural characteristics used.

Variable	Name	Min.	Max.	Mean	SD
Proportion (%) of saw-timber trees	Timber	19.9	43.7	33.5	0.06
Saw-timber diversity	Timber_div	0.72	0.90	0.80	0.03
Proportion (%) of productive forest land	Forestland	79.5	99.2	94.1	0.04
Proportion (%) of forests < 40 years	Young	32.2	51.5	41.3	0.05
Proportion (%) of forests > 120 years	Oldgrowth	2.9	9.2	5.3	0.01

inventories, satellite images, numerical base maps and contour maps (Tomppo et al., 1998; Ministry of Agriculture and Forestry, 2000).

We used the proportion of productive forest land from the three subcategories of forest land (see above) as an indicator of forest fertility and productivity at the regional scale (Table 2). Fertile forest types support high numbers of bird and beetle species, and a high diversity of red-listed forest-dwelling species (Nilsson, 1979; Sippola et al., 2002; Similä et al., 2006). Low productivity forest land comprises semi-open and oligotrophic mire ecosystems while waste land includes almost treeless habitats such as crags and open mires (Finnish Statistical Yearbook of Forestry, 2005).

The percentage of saw-timber trees was used to provide information on the structural characteristics of forests in the different observation units. The proportion of saw-timber trees was calculated from the sums of saw-timber tree volumes (m³) for the tree species mentioned below in relation to the total volume of growing stock (m³) per observation unit (Table 2). The technical terminology in inventories defines a saw-timber tree as one that has a minimum diameter at breast height of 15 cm, 16 cm or 18 cm for Scots pine, *Pinus sylvestris* L., Norway spruce, *Picea abies* (L.) H. Karst. or deciduous trees (including *Betula pendula* L. and *Betula pubescens* Ehrh.), respectively (Finnish Forest Research Institute, 2003). Many wood inhabiting and saproxylic species require such trees in the forest landscape (Heliövaara et al., 2004).

It has been suggested that the amount of mixed mature forest in an area can act as a structural indicator of biologically diverse boreal forests (Larsson, 2001). However, accurate statistics on the proportion of mixed mature forests are not available at the LAU-1 scale. Therefore, the regional diversity of four saw-timber tree species was described using the Shannon diversity index (Table 2). The index was calculated by using separate proportions of saw-timber volumes for spruce, pine, birch and the combined proportion of volumes of other deciduous tree species in the total growing stock. The index was calculated as follows:

$$H' = - \sum_{i=1}^S p_i \ln p_i \quad (1)$$

where p_i is the proportion of each tree species from total volume of species i and S is the number of species.

Table 3

The correlation coefficients for species variables.

	Ptevol	WRI_total	WRI_grouse	WRI_large	WRI_small	WRI_ungu	WRI_other
Ptevol	1	−0.07	0.05	−0.34**	0.09	0.16*	0.20
WRI_total		1	0.33**	0.58***	0.29*	0.40**	0.11*
WRI_grouse			1	0.25*	−0.32**	0.0014	0.22*
WRI_large				1	0.02	−0.16*	0.08
WRI_small					1	−0.001	0.27
WRI_ungu						1	−0.24*
WRI_other							1

* p -Value, significant 0.05–0.01.** p -Value, significant 0.01–0.001.*** p -Value, significant <0.001.

The age structure of forests is indicative of management and the natural disturbance history of the region. Tree age structure data was available as percentage data for nine different age classes of forest land at the LAU-1 scale. The age class of a forest stand (sample plot) is estimated from the dominant tree species. The nine tree age class groups were: open forest stand, 1–20 years, 21–40 years, 41–60 years, 61–80 years, 81–100 years, 101–120 years, 121–140 years and older than 140 years. Two combined age classes were used in our subsequent analyses: the proportion of forests younger than 40 years and the proportion of forests older than 120 years of age (Table 2). These age classes are indicative of forestry activity as well as the composition of old forests at the regional scale.

2.4. Statistical analyses

The abundance of the Siberian flying squirrel, the values of WRI-total, and WRI of five game animal groups were compared with each other using Spearman rank correlation coefficients. We used indirect ordination analyses to compare the species community compositions of different regions and to relate the wildlife communities to forest structural characteristics (Shaw, 2003; Crawley, 2007). In order to explore the general patterns of the dataset and the interrelationships between species data and forest structural variables, we performed a Euclidean distance-based, unconstrained principal component analysis (PCA) with standardized variables. The PCA used ordinary, unweighted linear regression and unweighted singular value decomposition (SVD) by default (Oksanen, 2008a). Statistical analyses were performed using the R software package version 2.6.1 with the vegan library, version 1.8–8 (The R Development Core Team, 2007; Oksanen, 2008a,b).

Seven species variables were chosen for use in the PCA: (1) relative abundance of the Siberian flying squirrel; (2) WRI for all 16 species combined; (3) WRI for large carnivores; (4) WRI for small carnivores; (5) WRI for grouse; (6) WRI for ungulates; and (7) WRI for other species. Five forest structural variables were used: (1) the proportion of productive forest land; (2) the proportion of saw-timber trees; (3) Shannon's diversity index for saw-timber trees; (4) the proportion of forests younger than 40 years; and (5) the proportion of forests older than 120 years. We plotted the results to demonstrate co-occurrence and determine any informative

patterns within the variables and the 39 observation units. Finally, we correlated the informative patterns identified by PCA with socio-economic and land-use variables. These variables were used in our earlier work as social and economic indicators of sustainable forestry (Myllyviita, 2008; Vehkamäki et al., 2006; Vierikko et al., 2008).

3. Results

Correlations between information obtained from different species were generally low (Table 3). As could be expected, there were highly significant correlations between the WRI for all 16 game animals combined and the five wildlife subgroups (i.e. components of the WRI). The overall WRI was highly correlated with those of large carnivores and ungulates, and less well correlated with those of grouse, small carnivores and other species (mountain hare and red squirrel). Low correlations between the various WRIs for wildlife groups could be explained by the high variability in the information, especially with respect to WRIs for ungulates and large carnivores. The WRI for grouse exhibited a slight positive correlation with WRIs for large carnivores and other species. Abundance of the vulnerable Siberian flying squirrel was negatively correlated with WRI for large carnivores, but exhibited only a slight positive correlation with WRI for ungulates (moose, roe deer, wild forest reindeer and white-tailed deer). Furthermore, there were negative correlations between the abundances of potential predator–prey pairs: between WRIs for small carnivores and grouse, and between WRIs for large carnivores and ungulates (Table 3).

The seven species/species groups and five forest structural variables produced eleven principal component (PC) axes. We present results of the first four axes, the eigenvalues of which were higher than 1.0, based on the broken stick model (Shaw, 2003). These four PC axes explained 77% of the total variation (Table 4). In addition, we present site scores for four axes in order to examine regional patterns identified by the principal component analysis (Table 5).

The first axis explained 32% of the total variation. The relative abundance of the Siberian flying squirrel (Ptevol) and grouse species (WRI_grouse), saw-timber tree diversity (Timber_div) and

Table 4
Unconstrained PCA showing variable loading values for the first four PC axes. Loading values higher than 0.5 or lower than −0.5 are in bold type.

Variable	Axes			
	PC1	PC2	PC3	PC4
Ptevol	0.779	−0.286	− 0.562	−0.417
WRI_total	0.387	− 0.539	1.096	−0.317
WRI_grouse	0.639	−0.496	0.340	0.513
WRI_large	−0.101	− 0.910	0.625	0.454
WRI_small	−0.222	−0.281	0.307	− 1.110
WRI_ungu	0.411	0.715	0.794	−0.434
WRI_other	0.217	− 0.673	−0.394	− 0.666
Timber	− 1.128	0.292	0.139	−0.264
Timber_div	0.738	− 0.552	− 0.529	−0.168
Forestland	− 1.152	−0.098	−0.125	−0.277
Young	− 1.027	− 0.695	−0.129	0.103
Oldgrowth	1.089	0.327	−0.039	−0.133
Eigenvalue	3.83	1.95	1.81	1.57
Cumulative variance	0.32	0.48	0.63	0.77

Note: Ptevol, relative abundance of the Siberian flying squirrel; WRI_total, abundance of 16 animal species combined; WRI_grouse, abundance of grouse, WRI_large, abundance of large carnivores; WRI_small, abundance of small carnivores, WRI_ungu, abundance of ungulates; WRI_other, abundances of the red squirrel and the mountain hare; Timber, the proportion of saw-timber trees from the volume of growing stock; Timber_div, saw-timber tree diversity; Forestland, the proportion of productive forest land; Young, the proportion of forests younger than 40 years; Oldgrowth, the proportion of forests older than 120 years.

Table 5

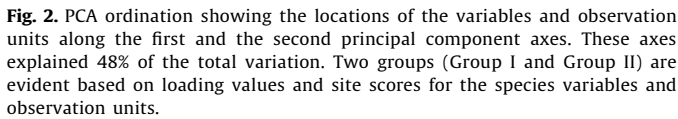
Site scores for the 39 observation units on the first four PC axes. Loading values higher than 0.5 are in bold type.

Code	Observation unit	PC1	PC2	PC3	PC4
41	Rauma	0.118	0.949	1.229	−0.913
43	Pori	0.197	1.135	0.612	−0.786
44	Pohjois-Satakunta	0.567	0.759	0.080	−0.169
61	Luoteis-Pirkanmaa	0.217	0.300	0.054	−0.637
62	Kaakkois-Pirkanmaa	−0.515	1.322	−1.201	0.424
63	Etelä-Pirkanmaa	−0.135	−0.009	1.094	−1.394
64	Tampere	−0.127	0.360	1.018	−1.523
68	Lounais-Pirkanmaa	−0.077	0.491	−0.102	−0.965
69	Ylä-Pirkanmaa	0.160	0.279	0.913	−0.172
101	Mikkeli	−1.132	0.479	−0.064	−0.127
102	Juva	−0.960	0.388	0.106	1.019
103	Savonlinna	−0.954	−0.885	0.803	0.043
105	Pieksämäki	−0.836	0.454	−0.627	−0.270
111	Ylä-Savo	−0.119	−1.252	−0.726	−0.439
112	Kuopio	−0.765	−0.299	0.383	−0.293
113	Koillis-Savo	−0.382	−0.945	0.165	0.598
114	Varkaus	−0.977	−0.070	0.036	−0.309
115	Sisä-Savo	−0.877	−0.290	−0.343	−0.167
122	Joensuu	−0.208	−1.289	1.045	1.123
124	Keski-Karjala	−0.826	−0.476	−0.130	0.315
125	Pielisen Karjala	0.204	−1.097	1.099	0.950
131	Jyväskylä	−0.814	−0.214	−0.537	−0.269
132	Joutsa	−1.001	0.472	−1.218	0.388
133	Keuruu	−0.449	−0.240	−0.285	0.227
134	Jämsä	−0.428	−0.049	−0.306	0.218
135	Äänekoski	−0.780	0.292	−0.253	0.947
138	Saarijärvi-Viitasaari	−0.272	−0.865	0.149	0.093
141	Suupohja	1.188	−0.253	0.217	0.571
142	Seinäjohti	1.068	0.360	0.001	−0.216
143	Eteläiset seinänaapurit	0.639	1.030	−0.656	0.963
144	Kuusioikkunat	0.593	1.193	0.891	0.455
145	Härmänmaa	0.902	−0.859	−0.967	−1.933
146	Järviselä	0.830	0.970	0.048	0.818
151	Kyrönmaa	1.034	−0.044	−1.762	0.582
152	Vaasa	0.999	−1.097	−0.633	−0.500
153	Syösterbottens kustregion	1.028	−0.865	−0.603	0.218
154	Jakobstadsregionen	1.036	−0.945	−0.512	−0.426
161	Kaustinen	1.149	−0.095	1.463	1.387
162	Kokkola	0.703	0.909	−0.485	0.170

the proportion of old forests (Oldgrowth) all had high positive loading values on the first PC axis (Table 4). The proportions of saw-timber trees (Timber), productive forest land (Forestland) and forests younger than 40 years of age (Young) all had low negative loadings. The first axis reflected variables that we considered to indicate good ecological conditions in boreal forests. Thirteen observation units received site scores higher than 0.5 on the first axis (Table 5). These observation units were closely associated with each other and were all located in the western part of the study area.

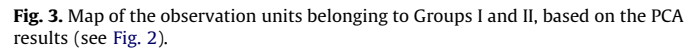
The second axis explained 16% of the total variation. WRI-ungulates was the only variable with a strong positive loading on this axis, while WRI-total, WRI-large carnivores, WRI-other, saw-timber tree diversity and the proportion of young tree age classes had high negative loadings (Table 4). Eight observation units had site scores higher than 0.5 on the second axis (Table 5). The second axis mainly represented the observation units characterized by a relatively high abundance of ungulates, taking no account of structural characteristics indicating good ecological conditions. These observation units were located in the south-western part of the study area.

The third axis explained 15% of the total variation. The variable WRI-total had a high positive loading on this axis. Surprisingly, both groups in the predator–prey pair large carnivores and ungulates received high positive loadings while the Siberian flying squirrel had a strong negative loading. Saw-timber tree diversity also had a strong negative loading. Ten observation units received site scores higher than 0.5 on the third axis. They were located in



The fourth axis explained 14% of the total variation. The variables WRI-grouse and WRI-other received the highest positive loading values, while WRI-small carnivores had a strong negative loading. Nine observation units had site scores higher than 0.5 (Table 5). The fourth axis mainly described observation units with a high relative abundance of grouse. These observation units were sparsely spread out across the whole study area. Because the third and fourth axes offered little information from the perspective of sustainable forestry, we decided to exclude these two axes from further analyses, even though they were informative from the context of wildlife species abundance.

The two groups were geographically distinct from each other (Fig. 3). The observation units in group I are restricted to the western part and the coastal region of the study area, where an agrarian culture dominates and where the remaining forests are characterized by mires of low productivity (Finnish Statistical Yearbook of Forestry, 2005). The units in group II are situated in the central and eastern parts of the study area, which is sparsely populated, on average financially poorer, and where local communities are more dependent on forestry than in the western part of the study area (Statistics Finland, 2005). The observation



The first principal component axis was most informative from an ecological perspective; we, therefore, correlated PC1 site scores with additional land-use and socio-economic variables (Fig. 4). We identified a strong positive correlation between the proportion of private forests owned by farmers (Agrifor) and scores on the first principal axis (PC1), while the proportion of other private forest owners (Private) and forest companies (Companies) had a strong negative correlation (Fig. 4). The proportion of non-industrial forest owners, such as the State, was not correlated with PC1 site scores. There was a slight positive correlation between PC1 site scores and gross regional forestry production (Forestry). No correlations were found between gross regional agricultural production (Agriculture), the total amount of designated conservation area (Total_reserve), the amount of old-growth forest reserves (oldgrowth_reserve), total gross regional production (Regiacc), the number of inhabitants and PC1 site scores (Fig. 4).

4.1. Associations in abundances of wildlife species and other variables

In units with structural characteristics that we considered reflecting good ecological conditions (i.e. high proportion of old

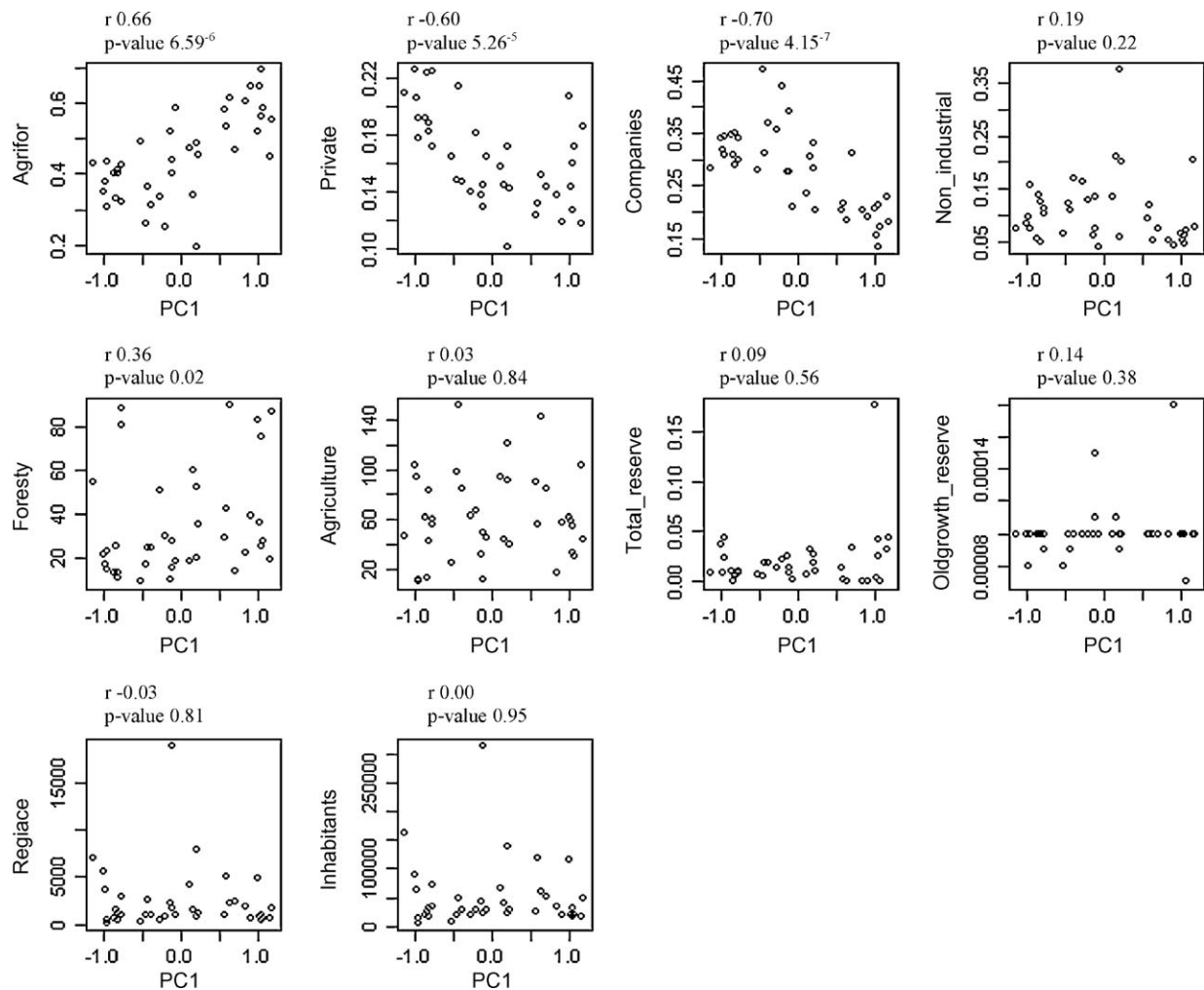


Fig. 4. Correlations between the site scores from PC axis 1 and socio-cultural and land-use variables (Myllyviita, 2008). Site scores belonging to group I (see Fig. 2) are situated to the right of the x-axis. *Note:* Agrifor, proportion of forest owned by private farmers; Private, proportion of forest owned by other private owners; Companies, proportion of forest owned by forest companies; Non_industrial, proportion of forest owned by State and other public owners (municipality, parish); Forestry, gross regional forestry production in 2005; Agriculture, gross regional agricultural production in 2005; Total_reserve, proportion of conservation areas on forestry land; Oldgrowth_reserve, proportion of permanent old-growth forest reserves on forestry land; Regiacc, gross regional production of industry, services, forestry and agriculture in 2005; Inhabitants, number of inhabitants in 2005.

forests, high saw-timber tree diversity), high grouse richness coincided with abundant Siberian flying squirrels. The forest structural data used, however, do not indicate whether the remaining old forests have been managed recently. Managed old forests differ from natural old-growth forests with respect, for example, to the amount of decaying wood and the tree species composition (Mykrä et al., 2000). The average amount of dead and decaying wood, however, increases with forest age (Ihalainen and Siitonen, 2006).

Both the flying squirrel and grouse (capercaillie, black grouse, hazel grouse and willow grouse) have suffered as a result of modern forestry activities (Hanski, 1998, 2006; Pellikka et al., 2006). An association between the abundance of flying squirrel and of grouse suggests that conservation management to support the flying squirrel, which has a high political conservation status, would also benefit grouse. On the other hand, as a highly valued game species, the habitat requirements of grouse are often taken into consideration in forestry practices which would benefit flying squirrels and other forest-dwelling species with similar habitat requirements. It is worth noting that neither the flying squirrel nor grouse necessarily need decaying wood for their survival. Therefore, the abundance of flying squirrel or grouse does not indicate a

large amount of coarse woody debris and the presence of specialized wood inhabiting species (but see Hurme et al., 2008).

Interestingly, forest ownership type correlated more strongly with the occurrence of the flying squirrel, grouse and diverse forest structural characteristics than did the total area designated for conservation. In Finland, forest management practices are commonly operationalized at the forest holding scale (Kurttila et al., 2002). It has been suggested that forest ownership diversity, which results in different management activities, can lead to increased ecological sustainability (Vehkamäki, 1997; Uutera et al., 1998; Spies and Johnson, 2003). The forest holdings in the western part of the study area are, on average, small (Finnish Statistical Yearbook of Forestry, 2005), resulting in a mosaic-like forest landscape with fine-grained forest structure supporting high abundance of flying squirrel and grouse.

The richness of small carnivores was more closely linked to the proportion of young forests than of old forests, and was negatively correlated with grouse abundance. Earlier studies have demonstrated that small carnivores and ungulates can benefit from a fragmented forested landscape, while grouse seem to suffer when there is fragmentation (Kurki et al., 1998, 2000). Small carnivores and ungulates (especially white-tailed deer and roe deer) also

benefit from an increasing amount of agricultural land, and moose in particular from the presence of tree seedlings (Härkönen, 1998; Nikula et al., 2004; Pellikka et al., 2006). Indices of small carnivore and ungulate richness could be considered potential indicators of the cumulative effects of human-created disturbances. In our study, relatively high ungulate abundance was found in the south-western part of the study area, which is characterized by an agrarian and fragmented forest landscape.

4.2. Regional distribution of variables

To discuss our results in the context of regional land-use history and biogeographical factors, two distinct regions were identified in terms of wildlife species abundance and forest structural characteristics. The western part of the study area, i.e., observation units belonging to group I, represented a complementary ('win-win') situation of sustainable forestry with indicators of relatively good ecological and socio-economic conditions. Boreal forests in this region were, on average, less diverse with respect to forest species compared to the middle-eastern part of the study area, a result of both geography and a long and intensive forest utilization history in the west (Punttila, 2000; Virkkala and Rajasärkkä, 2000; Vehkamäki, 2006). The presence of flying squirrels in the western part of the study area reflects, at some level, forest patch connectivity. The dispersal ability of the flying squirrel, however, has been shown to be good and dispersal distances can be long in forested landscapes in southern Finland (Selonen and Hanski, 2004; Hanski and Selonen, 2009).

The landscape in the observation units belonging to group I is characterized by agricultural land and large, semi-open mires; any old forests are mostly isolated and fragmented (Vierikko et al., 2008). For some forest-dwelling species, these old forest patches among pine-dominated mires do not offer suitable habitats, and the landscape ceases to be suitable for species with restricted dispersal ability (Mykrä et al., 2000; Kurttila et al., 2002). In addition, the total volume of CWD, and the abundances of threatened forest-dwelling species, such as beetles and birds, has been shown to be an average lower in conservation areas in the west than in the eastern part of the study area (Punttila, 2000; Virkkala and Rajasärkkä, 2000; Finnish Statistical Yearbook of Forestry, 2005).

In the middle-eastern part of the study area (group II) the proportions of young forests, saw-timber trees and productive forest land were closely linked, indicating economically sustainable forestry and strong management pressures. In general, the boreal forests in these areas are more productive and diverse than those in the west, with relatively high proportions of herb-rich forests and viable populations of many endangered forest species (Virkkala and Rajasärkkä, 2000; Kallio et al., 2008). In the group II region forest holdings are, on average, larger than in the western part of the study area (Finnish Statistical Yearbook of Forestry, 2005). Private large forest holdings seemed to be managed more actively than small forest holdings; this can reduce the mosaic-like and fine-grained forest structure. A forest landscape with large young forest patches and homogeneous forest structure is not well-suited to flying squirrel and species of grouse.

5. Conclusion

We found close association between the abundance of wildlife species and the structural characteristics considered to indicate good ecological conditions in boreal forests. In contrast, the proportion of young forests (indicative of high management pressure) was large in regions where forests are generally diverse and productive. Interestingly, the proportion of forests owned by farmers was closely and positively associated with the abundances

of the Siberian flying squirrel and grouse, and with the presence of old forests and high saw-timber tree species diversity. However, the extent of conservation areas (an indicator of sustainable forestry) did not correlate with the abundance of grouse or flying squirrel.

As mentioned earlier, indicators are used to monitor forest policy goals and to provide valuable information for decision-makers. Indicator use, however, always depends on the context in which it is applied: biogeographical, land-use history and socio-cultural factors have to be taken into consideration when making regional comparisons. When discussing results from these perspectives we noticed that 'ecologically sustainable' regions were located in observation units with a long history of intensive land-use, and with a lot of mires and low-productivity forest land. In these regions, sustainable forestry practiced mostly by farmers has, so far, supported the survival of some forest-dwelling species. In contrast, relatively low abundances of grouse and flying squirrel were found in regions with indications of intense forestry practiced by other private owners and forest companies. In these regions forests were generally more diverse and productive creating a conflict between ecological and economic aspects of forestry.

We feel that a 'shopping basket' containing a variety of taxonomic groups, forest structural characteristics and socio-economic factors together could be used as a valuable indicator tool for monitoring sustainable forest policy goals and the relationships between the ecological, economic and social dimensions (Niemelä and Baur, 1998). To conclude, we emphasize that our method must be implemented with caution, and more studies are needed to develop an accurate 'shopping-basket' for sustainable forestry. This study, however, has convinced us that studying a combination of indicators is useful in developing and refining explicit indicators for sustainable forestry.

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